A-A-54019
13 November 1989

COMMERCIAL ITEM DESCRIPTION

KNIFE, COMPOUND, DENTAL (with Detachable Blade)

The General Services Administration has authorized the use of this Commercial Item Description.

This commercial item description covers a dental compound knife with detachable blades suitable for trimming dental impression compound.

Salient characteristics.

Material.

Handle. The handle shall be fabricated of suitable colored phenolic material, conforming to ASTM Designation D700.

Handle nose. The nose shall be fabricated of corrosion-resisting steel.

Blades. The blades shall be fabricated of a carbon steel having a carbon content of 1.20 percent minimum. The steel shall have a uniform, fine grain structure.

Design and dimensions. The design and dimensions of the handle, handle nose and blade shall conform to figures 1 and 2. The handle proper and handle nose shall be permanently secured to each other in proper alignment.

Blades. The blades shall be hardened and tempered, and of one-piece construction. Blades shall fit securely and shall be easily detachable from the handle.

Tang. The tang shall comprise that part of the knife blade which fits the knife handle. It shall have a perforation and an angular rear edge which shall comply with the design and dimensions shown in figure 1. Dimensions not shown shall be in proportion with those indicated in the figures.

AMSC N/A FSC 6520

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Sharpness and cutting edge angle. The point end and the cutting edge shall be surgically sharp. The cutting edge shall have an included angle not greater than 36 degrees. The two angles forming the included angle shall be 13 ± 5 degrees.

Stiffness. Blades shall be rib-reinforced (see figure 2) and shall have sufficient stiffness not to deflect sideways beyond the torque limits of 0.96 inch-pounds under any conditions of normal use. The tang shall have sufficient resistance to lateral deflection to obviate the possibility of accidental separation of the blade from the handle. When blade is tested as specified herein, in no case shall the applied torque deflect the movable arm through an angle of more than 14 degrees for the point or the tang.

Hardness. Hardness of the finished blade shall be Rockwell 30N 75.5 to 83 when tested as specified in ASTM Designation E18.

Finish. The plastic handles shall have a smooth finish, except for the hand gripping area which shall have a textured finish. The handle nose shall have a satin finish. The blades cutting edge shall be smooth, except for sharpened cutting edge which shall have a fine honed finish.

Identification marking. Each blade shall be permanently and legibly marked with No. 25 and each handle shall be permanently and legibly marked with No. 6. Each component shall be permanently and legibly marked with the name or registered trademark of the manufacturer.

Workmanship. The handle shall show no cracks, or become soft or sticky. The handle nose shall have no pits, cracks, burrs or feathers. The blades shall have no pits, burrs, or feathers. The assembled compound knife shall be free from any other defects which detract from its appearance or may impair its serviceability.

Unit. Package (PG). One package containing one handle, and six blades, as specified, constitutes one unit. Each blade, shall be on a strip or individually packaged.

Unit package. Each handle and six blades on a strip or individually packaged shall be packaged in a durable bag or box.

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Quality Assurance Provisions

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of supplies to which such records relate.

Inspection. Inspection, as used in this specification, is defined as both examination (such as visual and auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Certificates of quality. Certificates of quality, supplied by the manufacturer of the component or materials, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government.

Sampling for inspection. Sampling for inspection shall be conducted in accordance with MIL-STD-105. Where feasible, the same sample shall be used for the determination of two or more inspection, examination and test characteristics. The unit of product for inspection, examination and test purposes shall be one handle (with handle nose) and one blade. Unless otherwise specified, sampling for examination shall be conducted at inspection level II, except for dimensions which shall be conducted at inspection level S-1 and sampling for tests (when applicable) shall be conducted at inspection level S-2.

Examination. Examination shall include, but shall not be limited to the following:

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Defects.

Handle

Handle and handle nose dimensions not as specified.
Finish not as specified.
Labels or printed marking not legible and not free from defacing marks, scratches and tears.

Blade

Blade (except tang) dimensions not as specified.
Tang not of design and dimensions specified.
Angle of cutting edge not as specified.
Cutting edge and point not central and uniform.

Examination under magnification. Blades shall be examined under ten power (10X) magnification for the following defects:

Defects

Cutting edge and point with feathers, nicks, burrs, or pits.
Point rounded, blunt, bent or broken.
Blade not smooth and not free from pits, burrs, grind marks, chips or breaks.

Blade shall be examined on all surfaces (flat, on edges and on ends).

Tests.

Handle. The handle shall be autoclaved for a period of at least 20 minutes at a temperature of 250 degrees Fahrenheit (^{0}F). Then the handles shall be allowed to cool to room temperature and examined for softness or stickiness.

Blade.

Stiffness. The stiffness of the blades shall be determined by use of a deflection test jig or similar device as specified herein.

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Testing device. The testing device shall be provided with two pairs of flat jaws, one pair which is fixed to hold the blade securely in a horizontal position. The other pair of jaws shall be provided with suitable weights, scales, and arm so that when this pair of jaws is clamped to the blade which is held by the fixed pair of jaws, the deflection shall be indicated for a particular torque applied to the blade.

Deflection of point end. The blade shall be clamped in the fixed pair of jaws with the pointed end projecting, and the center line of the perforation perpendicular to the edge of the jaws. The perforation shall be completely within the grip of the jaws, with the end of the perforation nearest the point of the blade within 0.0312 inch of the edge of the jaws. The other pair of the jaws shall be clamped to the blade 0.25 ± 0.0156 inch from the fixed jaws. The applicable torque specified herein, shall be applied to the blade through the movable jaws and the deflection noted to determine compliance with the requirements specified herein.

Deflection of tang end. The pointed end of the blade shall be clamped in the fixed pair of jaws with the center line of the perforation perpendicular to the edge of the jaws, and with 0.4375 ± 0.0156 inch of the perforation projecting beyond the edge of the jaws. The other pair of jaws shall be clamped to the blade 0.25 ± 0.0156 inch from the fixed jaws. A torque of 0.80 inch-pounds shall be applied to the blade through the movable jaws and the deflection noted to determine compliance with the requirements specified herein.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch/pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this document are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerance specified in the inch/pound units, a request should be made to the contracting officer to determine of the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

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Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Public Law 94-580, as amended, and as implemented by Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

NOTE: The following National Stock Number is covered by this document:

National Stock Number	Item Identification
6520-00-286-4307	KNIFE, COMPOUND, DENTAL, with Detachable Blade

Ordering data. Intermediate/exterior package quantities and labeling and marking must be as specified in the contract and/or order.

MILITARY INTERESTS:

Custodians:
Army - MD
Navy - MS
Air Force - 03

PREPARING ACTIVITY:

DoD-MB

Agent:

DLA-DM

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CIVIL AGENCY COORDINATING ACTIVITIES:
VA-OSS
PHS

PROJECT NO. 6520-2286

LOCATION: NCR Disk No. 7

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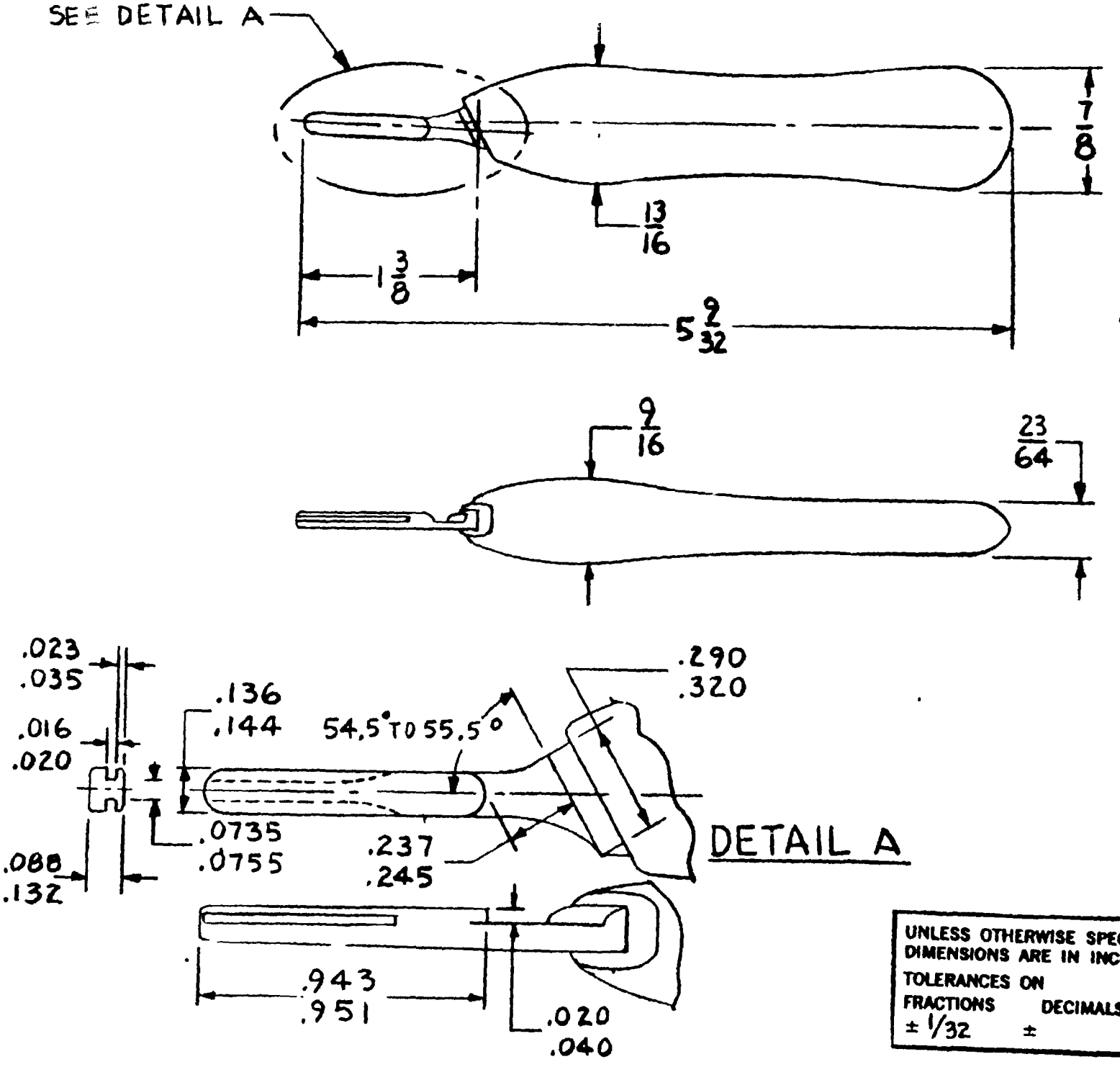


FIGURE 1 HANDLE FOR KNIFE, COMPOUND, DENTAL

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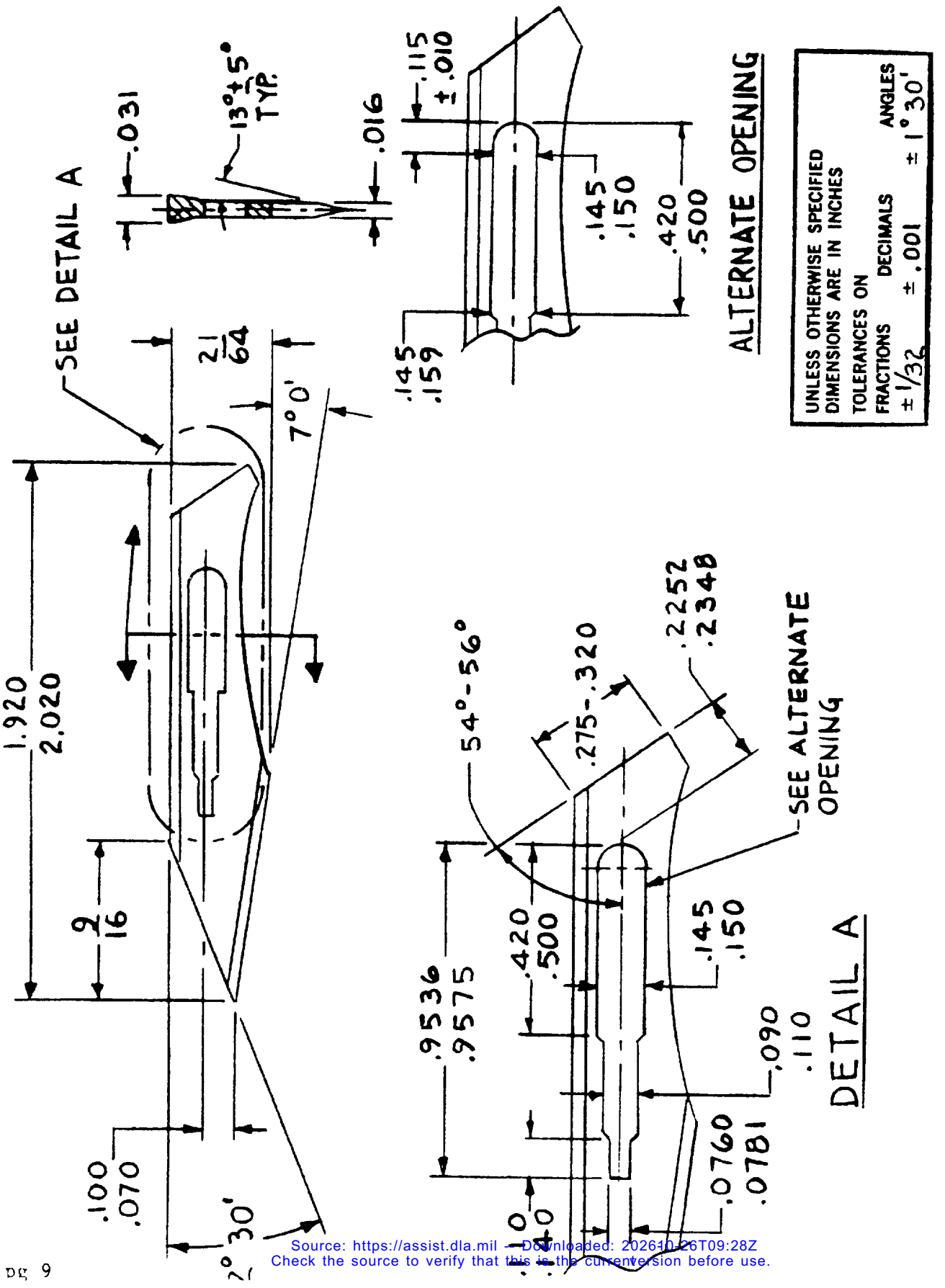


FIGURE 2 BLADE FOR KNIFE, COMPOUND, DENTAL

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